NMZ Serratia genomics

Glossary for the research group. Companion to the 21 October 2026 explainer.

Core terms

Term	Meaning and relevance
Isolate; strain	An organism recovered from a sample/culture; a distinguishable lineage. The isolate name is not itself a species identification.
Genotype; phenotype	Genetic constitution; an observed trait under stated conditions. Presence of a candidate gene does not predict its quantitative effect by itself.
Read; basecalling	A recorded sequence fragment; conversion of nanopore electrical signal into DNA bases.
Assembly; contig; polishing	Reconstructing longer sequences; a continuous assembled sequence; refining that reconstruction with read evidence.
Chromosome; plasmid; replicon	A main genomic DNA molecule; an additional replicating DNA element; the general term for a replication unit. Plasmid identity requires evidence.
bp, kb, Mb	Base pairs, thousand base pairs, million base pairs. They measure sequence length.
Annotation; CDS; predicted proteome	Inferred genome features/functions; a protein-coding sequence; the protein set predicted from annotation. Here FAA files also contain ten sORFs per isolate.
GC content	Percentage of DNA bases that are G or C. A descriptive property, not a degradation-performance score.
Gene; protein	The DNA sequence and its encoded product. Write the gene as italic *swrW* and the protein as SwrW in manuscript prose.
Homolog	A sequence related by shared ancestry. This does not guarantee the same substrate or function.
Pathway; substrate	A set of chemical steps; the molecule an enzyme or pathway acts on. Substrate identity matters when connecting genomics to an assay.
Biosynthetic gene cluster (BGC)	Nearby genes associated with making a specialized molecule. A software region may include several predictions; 11 regions do not mean 11 demonstrated compounds.
Biosurfactant; amphiphile	A biologically produced surface-active substance; a molecule with water-compatible and water-avoiding portions. Activity and molecular identity need distinct evidence.
Serrawettin W1; serratamolide; congener	W1 is also called serratamolide; congeners such as serratamolide C are related but chemically distinct variants. Do not count aliases as two independent discoveries.
EPS; CPS; capsule; biofilm	EPS here means extracellular polymeric substances: a matrix that may include polysaccharides, proteins and DNA. CPS means capsular polysaccharide, associated with the cell. Biofilm describes organized microbial growth often associated with a matrix. These terms do not identify one chemical product.

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PNAG	Poly-beta-1,6-N-acetylglucosamine, a matrix/adhesion polysaccharide associated with the *pga* system. A lone *pgaB* annotation does not establish the whole system.
Cellulose; bcs	A beta-1,4-linked glucose polymer; genes associated with bacterial cellulose synthesis, processing and export. The focal bcs loci support candidates, without measuring cellulose production.
Gene cluster; operon	Nearby genes implicated in a shared function; genes transcribed together as a unit. A predicted region or conserved neighborhood is not automatically an experimentally established operon.
Hydrocarbon; aromatic compound	A molecule containing only carbon and hydrogen; a compound with aromatic ring chemistry. Oxygenated aromatics can be aromatic without being hydrocarbons.
Alkane; oxidation; monooxygenase	A saturated hydrocarbon; a chemical oxidation step; an enzyme incorporating one oxygen atom into a substrate in the relevant reaction. An enzyme-family label does not identify the tested substrate.
Phylogeny; branch support	Inferred sequence relationships; support for a grouping under the analysis. A 99% supported node is not 99% certainty of biochemical activity.
Gene neighborhood; synteny	Local gene arrangement; conserved gene organization in the comparative usage here. Proximity does not prove co-regulation or a shared substrate.
LC-HRMS; MS/MS	Liquid chromatography-high-resolution mass spectrometry; fragmentation of selected ions to help distinguish structures. Exact mass alone may fit several compounds.
Tentative annotation; confirmed identity	A candidate chemical label; an identification supported by the appropriate reference/standard evidence. Genomics can support plausibility without upgrading analytical confidence.
Peak-area percentage	Relative detector response under an analysis/denominator. It is not automatically concentration, purity, mass fraction or yield.
Surface tension; interfacial tension	Physical properties at a liquid-gas or two-phase interface. Report the actual interface and measurement method.
Emulsification index	A protocol-dependent measure of emulsion formation/stability, often involving layer height after a defined time. It is not the same as tension reduction.
Pour point; PPD	A low-temperature flow endpoint under a defined test; pour-point depressant. A lower pour point does not by itself establish chemical degradation.
Transformation; mineralization	Chemical conversion of a substrate; conversion to inorganic end products. Apparent disappearance alone establishes neither.
Virulence-associated homolog	A match to a protein implicated in a pathogenic process in a reference context. It does not measure this isolate's pathogenicity or establish safety.

Technical terms - optional reading

Term	Meaning and relevance
ONT; WGS; FASTQ; FASTA	Oxford Nanopore Technologies; whole-genome sequencing; sequence plus per-base quality; sequence without those quality fields.
Q score; N50; sequencing depth	A logarithmic quality estimate; length above which half the bases occur in sequences that long or longer; mapped read representation per position. None alone proves assembly accuracy.
ANI; AF; dDDH	Average nucleotide identity across aligned genome regions; fraction of sequence aligned under the method; computational estimate of DNA-DNA hybridization relatedness. ANI is not identity across every base of the whole genome. Species boundaries depend on methods and references.
GTDB; GTDB-Tk; TYGS	Genome Taxonomy Database; its local classification toolkit; Type (Strain) Genome Server, a web service using type-strain comparisons, GBDP distances, dDDH and a phylogram. They are systems that use relatedness evidence, not names of equivalent metrics.
ANI screening; species radius	Classification against reference genomes using ANI and aligned fraction; the representative-specific identity boundary used by GTDB. The focal queries were classified by screening, without marker-tree placement.
GBDP	Genome BLAST Distance Phylogeny: genome-comparison methods used by TYGS. The d4 formula for the focal dDDH estimates is distinct from the d5 distances used for the tree.
Type strain; representative; LPSN	Strain anchoring a species name; database comparison genome; nomenclature resource. A representative need not be the type strain.
Ortholog; paralog; orthogroup	Homologs related through speciation; through duplication; an inferred group under a method's rules. Conserved function is not guaranteed.
ProteinOrtho; PoFF	Protein grouping method; its gene-order extension. Keep sequence and gene-order outputs distinct.
Identity; alignment coverage; E value	Exact residue matches over an alignment; fraction of a specified sequence covered; expected chance matches in the search space. State the denominator and threshold.
HMM; Pfam; MIBiG	Protein-family statistical profile; protein-family resource; characterized BGC reference database. Matching a model/reference is not an activity assay.
NRPS; C-A-PCP-TE	Nonribosomal peptide synthetase; condensation, adenylation, carrier and thioesterase domains. Modules can be used iteratively; domain labels alone do not determine the complete product.
hpa; paa; pob/pca	Labels for hydroxyphenylacetate, phenylacetate and hydroxybenzoate/protocatechuate-associated machinery in this interpretation. Use locus/product evidence rather than gene names alone.
4-HBA; PobA; PcaK	4-hydroxybenzoate; a hydroxylase acting on it in characterized systems; an aromatic-acid transporter. A downstream PobA-like hit does not establish the upstream steps of a complete toluene route.
wzb/wzc; HADEG Emulsan category	Generic capsule-associated phosphatase/kinase markers. Their database category does not determine the chemical structure or activity of a polymer produced by NMZ.
LadA; FMN; LLM	A monooxygenase-associated family name; flavin mononucleotide cofactor; luciferase-like monooxygenase terminology. Here LLM refers to an enzyme family.

Term	Meaning and relevance
AlkB naming ambiguity	DNA-repair AlkB and membrane alkane hydroxylase AlkB are different proteins. The focal annotations concern the former.
Adduct; mass error; authentic standard	An ion associated with another species such as H+ or Na+; measured versus expected mass difference; a reference compound measured for comparison.
ShlA-ShlB; Has; T6SS; VFDB	Hemolysin/secretion-activation pair; heme-acquisition system; type VI secretion system, often involved in competition; Virulence Factor Database. Multiple labels may match the same locus.
Causality; controls; replicate	Evidence that changing a factor changes an outcome; comparisons addressing alternative explanations; independently repeated units whose biological/technical level must be defined.